

1. Lab Overview

In this lab, I built a basic Windows domain environment using a **Windows Server 2019 virtual machine** and a **Windows 10 client virtual machine**. The main goal was to install **Active Directory Domain Services**, promote the Windows Server to a **domain controller**, create a domain, and connect a Windows 10 client computer to that domain.

This lab helped me practice real IT support and junior system administration tasks, including server setup, domain configuration, client connectivity, and Active Directory verification. These are important skills for roles such as **IT Support Analyst, Help Desk Analyst, Desktop Support Technician, and Junior System Administrator**.

2. Business Scenario

This lab simulates a small organization or college-style IT environment where an IT support team needs to create a centralized domain environment.

As the system administrator, I configured a Windows Server 2019 VM as the domain controller and joined a Windows 10 client VM to the domain. This setup allows users, computers, and resources to be managed centrally through Active Directory.

3. Lab Objectives

The objectives of this lab were to:

- Install Active Directory Domain Services.
- Configure communication between the Windows Server 2019 VM and Windows 10 VM.
- Rename the Windows Server 2019 VM.
- Promote the server to a domain controller.
- Create a new Active Directory forest/domain.
- Verify AD DS installation using Event Viewer.
- Rename the Windows 10 client VM.
- Join the Windows 10 client to the domain.
- Verify the client computer object in Active Directory Users and Computers.

Your original lab lists the main goals as installing AD DS and connecting a client system to the domain.

4. Lab Environment

Item	Details
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Server VM	Windows Server 2019
Client VM	Windows 10
Server Name	WARA-SERVER
Client Name	WARA-CLIENT
Domain Name	wara.edu
Main Role Installed	Active Directory Domain Services
Main Tools Used	Server Manager, Active Directory Users and Computers, Event Viewer, Command Prompt
Network Test	Ping between server and client
Firewall Rule	ICMPv4 / Ping Allowed

Before installing AD DS, both VMs were checked with `ipconfig` to confirm they were on the same network. ICMPv4 was also allowed so the systems could ping each other. The server was configured with a static IP address to prevent domain connectivity issues.

5. Tools Used

- Windows Server 2019
- Windows 10
- VMware
- Server Manager
- Active Directory Domain Services
- Active Directory Users and Computers
- Event Viewer
- Command Prompt
- Windows Firewall Advanced Settings

Step 1: Prepared the Virtual Machines

I started by confirming that the Windows Server 2019 VM and Windows 10 VM were on the same network. I used `ipconfig` on both machines and enabled ICMPv4 traffic by creating a firewall rule named **Ping Allowed**. This allowed the machines to ping each other and confirm network communication.

This step was important because the client computer needed network connectivity to communicate with the domain controller.

Step 2: Renamed the Windows Server 2019 VM

I renamed the Windows Server 2019 VM to:

WARA-SERVER

After changing the computer name, I restarted the server so the new name would apply.

Step 3: Installed Active Directory Domain Services

Using **Server Manager**, I installed the **Active Directory Domain Services** role on the Windows Server 2019 VM.

The process included:

1. Opening Server Manager.
2. Selecting **Manage > Add Roles and Features**.
3. Choosing **Role-based or feature-based installation**.
4. Selecting the server from the server pool.
5. Selecting **Active Directory Domain Services**.
6. Adding the required features.
7. Installing the role.

The lab instructions show that AD DS was installed through Server Manager using the Add Roles and Features wizard.

Confirm installation selections

DESTINATION SERVER
WARA-SERVER

Before You Begin

Installation Type

Server Selection

Server Roles

Features

AD DS

Confirmation

Results

To install the following roles, role services, or features on selected server, click Install.

☒ Restart the destination server automatically if required

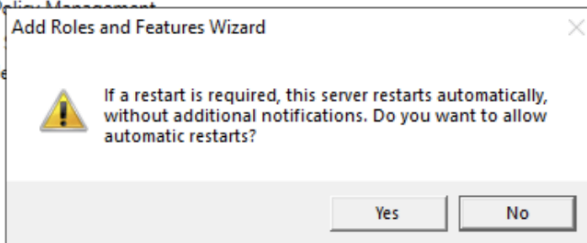
Optional features (such as administration tools) might be displayed on this page because they have been selected automatically. If you do not want to install these optional features, click Previous to clear their check boxes.

Active Directory Domain Services

Group Policy Management

RemoteApp and Desktop Connections

Role Services

[Export configuration settings](#)[Specify an alternate source path](#)

< Previous

Next >

Install

Cancel

Events

1 Events

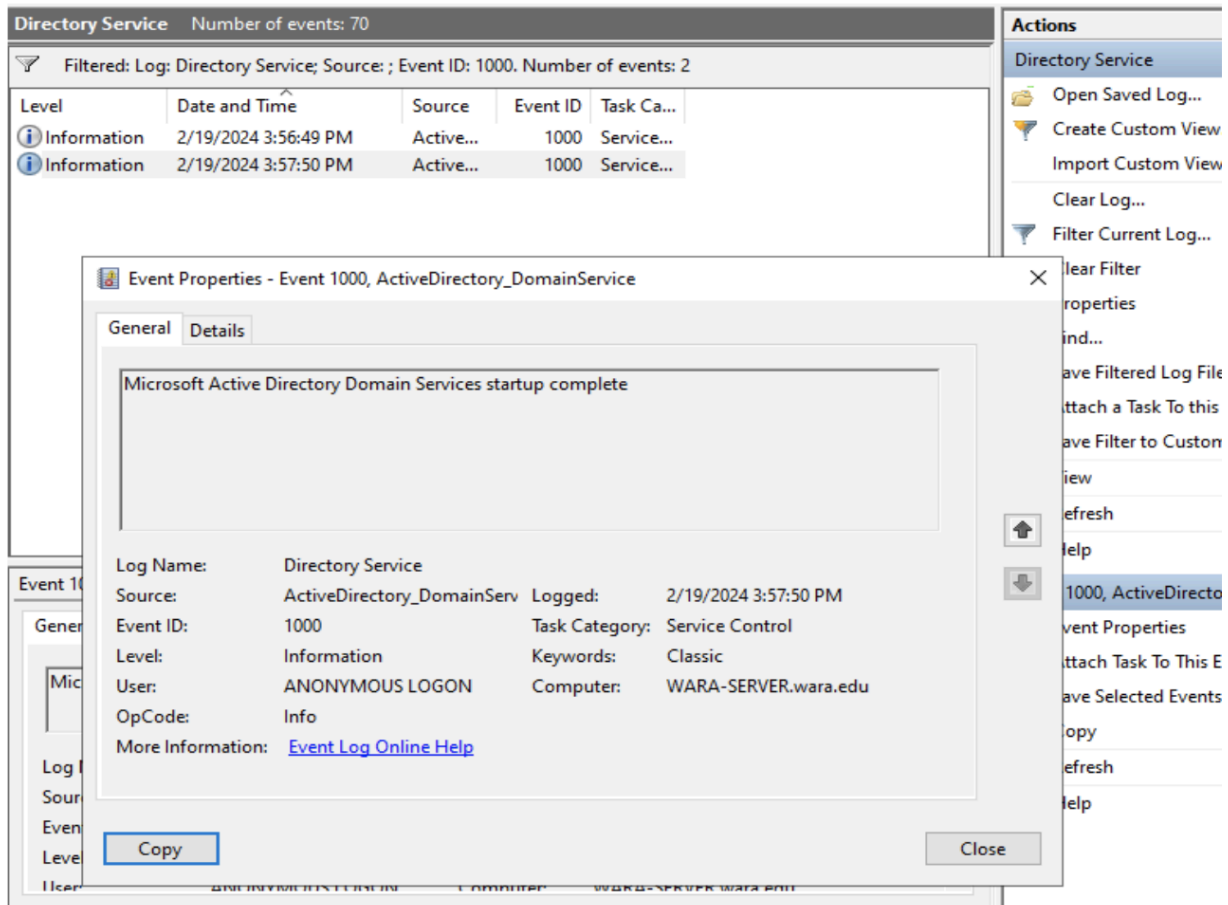


Figure 1: Active Directory Domain Services role installation on Windows Server 2019.

Step 4: Promoted the Server to a Domain Controller

After installing AD DS, I promoted the server to a domain controller.

During the configuration, I selected:

Add a new forest

The domain name used was:

wara.edu

I kept **DNS Server** and **Global Catalog** selected, entered the Directory Services Restore Mode password, reviewed the configuration, and completed the installation.

The original lab shows the server being promoted to a domain controller and a new forest being created using the domain name **wara.edu**.

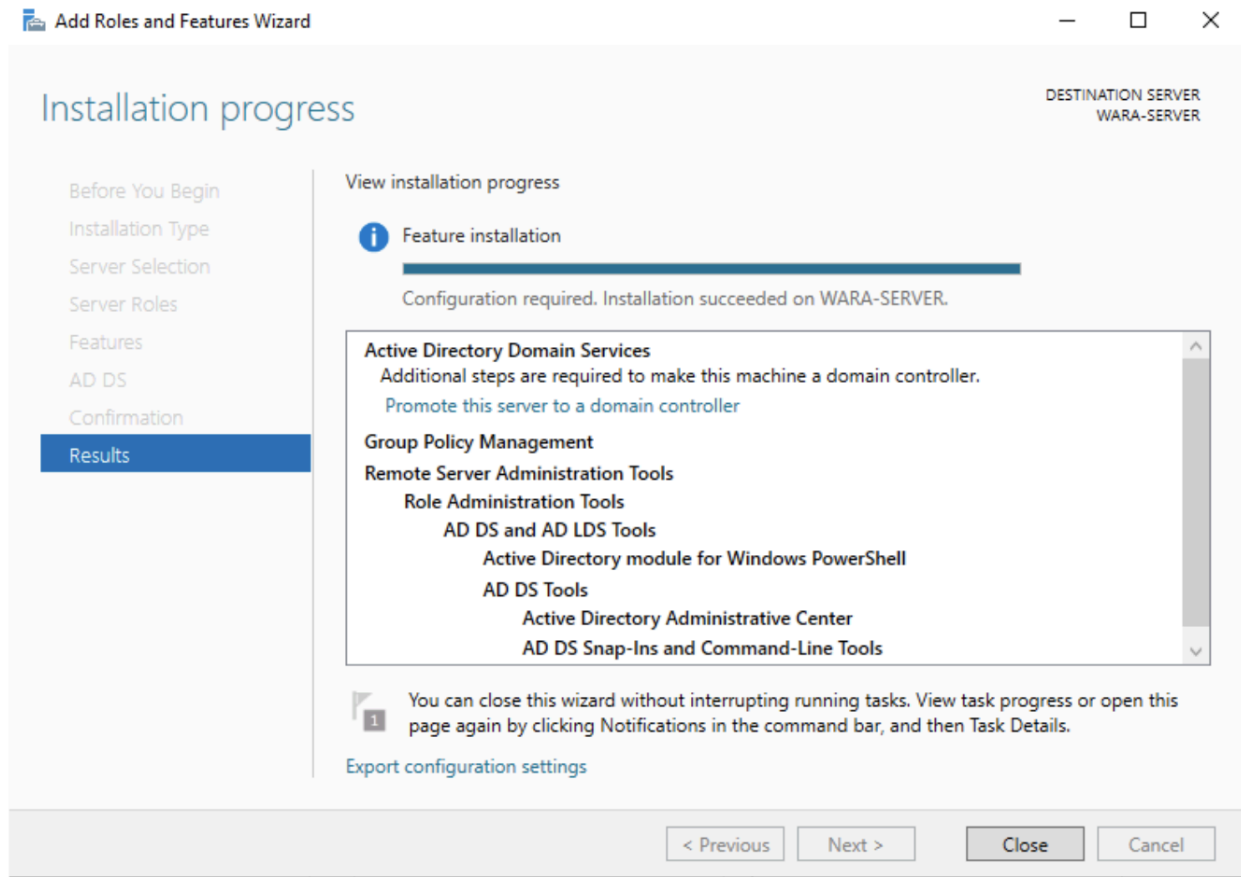


Figure 2: Windows Server 2019 promoted to a domain controller for the wara.edu domain.

Step 5: Verified AD DS Installation in Event Viewer

After the server restarted, I verified that Active Directory Domain Services started correctly.

I opened:

Server Manager > Tools > Event Viewer

Then I navigated to:

Applications and Services Logs > Directory Service

I filtered for **Event ID 1000**, which confirmed that Microsoft Active Directory Domain Services startup was completed.

The lab specifically used Event Viewer and Directory Service logs to verify AD DS role installation by filtering for Event ID 1000.

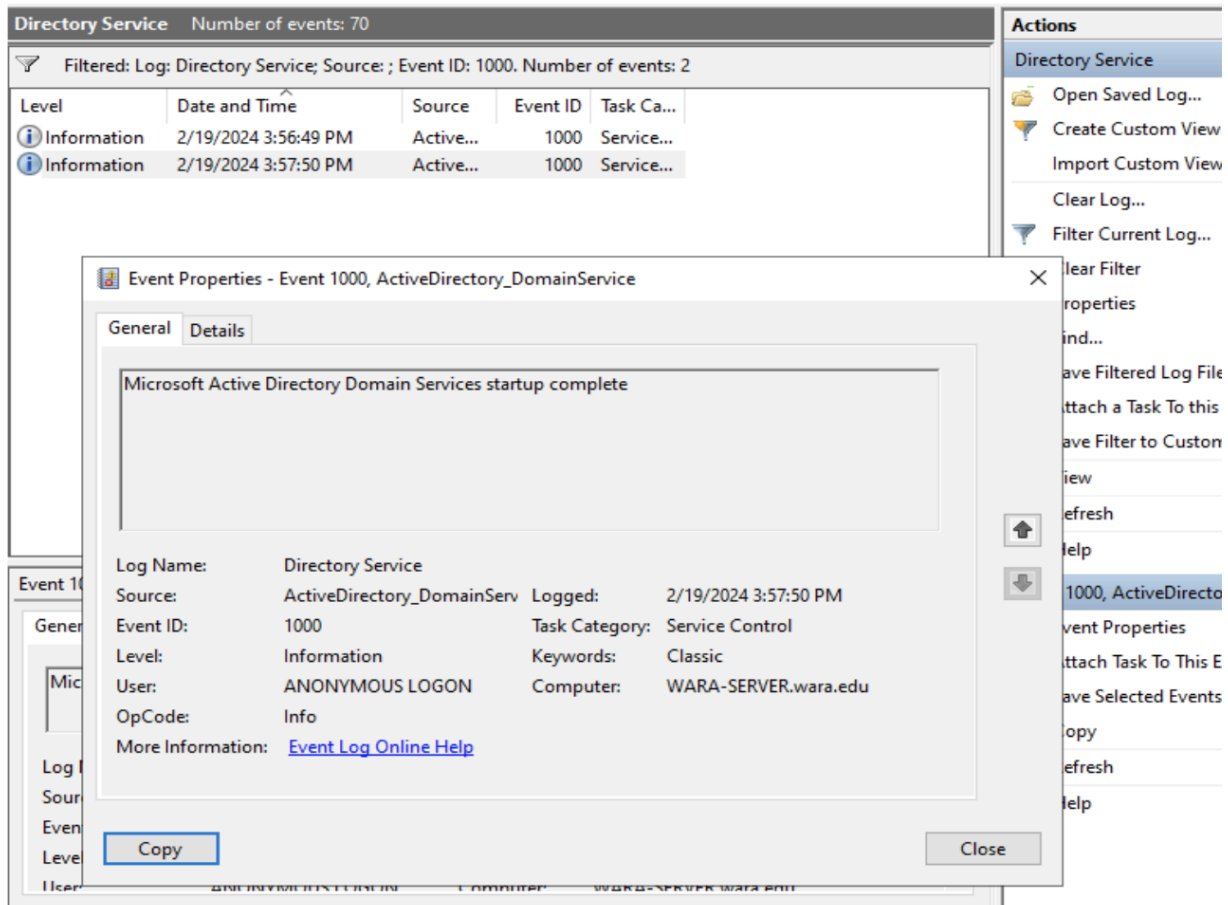


Figure 3: Event Viewer verification showing Active Directory Domain Services startup completed.

Step 6: Renamed the Windows 10 Client VM

Next, I renamed the Windows 10 client VM to:

WARA-CLIENT

After renaming the client computer, I restarted the VM so the name change would apply.

Step 7: Joined the Windows 10 Client to the Domain

After renaming the Windows 10 client, I joined it to the domain.

I opened the system properties, selected the domain option, and entered:

wara.edu

Then I authenticated using the domain administrator credentials. After the domain join completed, Windows displayed a welcome message and required a restart.

The lab explains that the Windows 10 VM was joined to the Active Directory domain by entering the domain name and authenticating with the Administrator account.

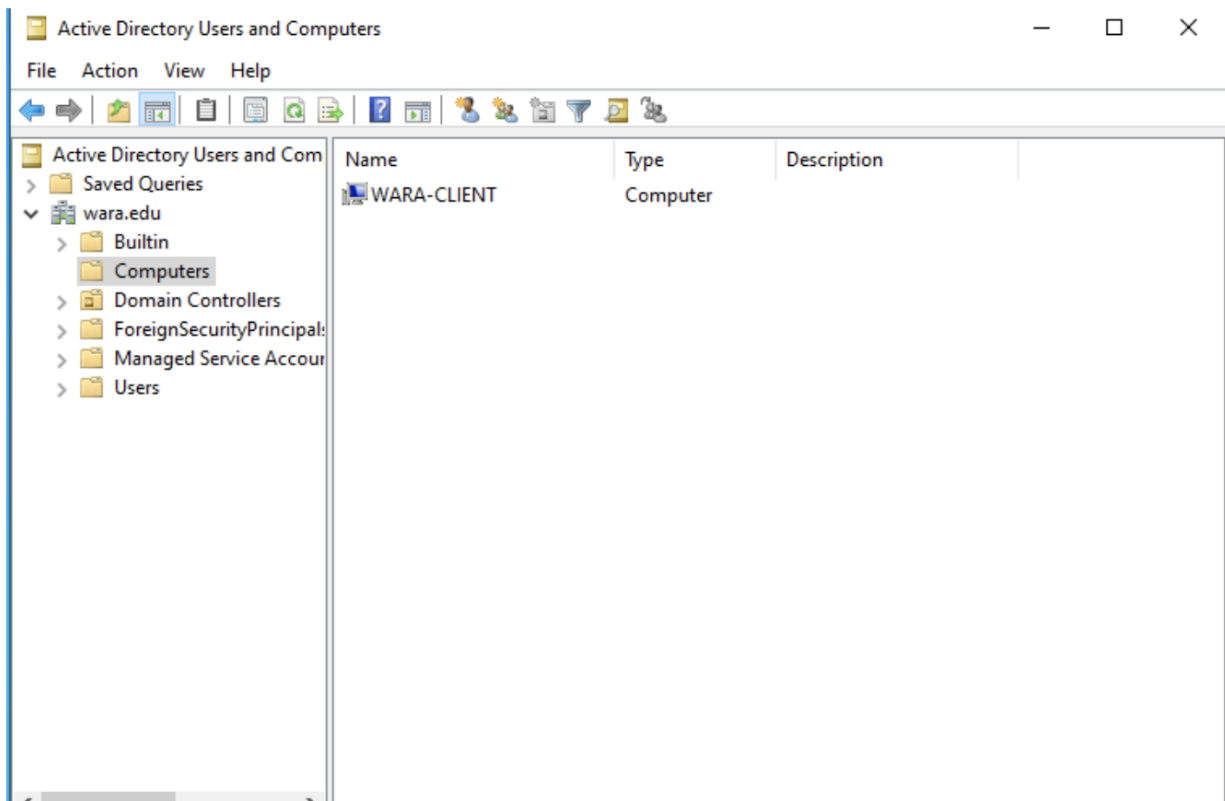


Figure 4: WARA-CLIENT successfully added to Active Directory after joining the domain.

7. Key Findings

Finding	Explanation
Server and client communication was required	The Windows Server and Windows 10 client needed network connectivity before domain setup.
Static IP is important for servers	A static IP helps prevent domain controller reachability problems.
AD DS installed successfully	The Windows Server 2019 VM was able to install the Active Directory Domain Services role.

Domain controller promotion was successful	The server was promoted and the wara.edu domain was created.
Event Viewer confirmed AD DS startup	Event ID 1000 showed that Active Directory Domain Services started successfully.
Client joined the domain	WARA-CLIENT appeared in Active Directory Users and Computers.

8. Analysis

This lab showed how a Windows domain environment is created from the ground up. Installing AD DS and promoting a server to a domain controller are important steps because they allow an organization to centrally manage users, computers, groups, and policies.

Joining the Windows 10 client to the domain confirmed that the domain controller was working and that the client could communicate with the server. This is a key part of real IT environments because most organizations use centralized identity and device management.

For IT support, this lab is important because many support issues involve domain logins, computer accounts, authentication, DNS, and connectivity to domain resources.

9. IT Support Relevance

This lab is directly related to IT Support Analyst and Junior System Administrator work.

It connects to real job tasks such as:

- Joining computers to a domain
- Verifying computer objects in Active Directory
- Troubleshooting domain connectivity
- Understanding domain controller setup
- Checking Event Viewer logs
- Supporting Windows clients in a business network
- Understanding why DNS and static IPs matter for domain services
- Helping users and devices authenticate in a domain environment

10. Skills Demonstrated

This lab demonstrates:

- Windows Server 2019 administration
- Active Directory Domain Services installation
- Domain controller promotion
- Domain creation
- Windows 10 domain join
- Client/server connectivity testing
- Event Viewer verification
- Active Directory Users and Computers
- Basic network troubleshooting
- Technical documentation

11. Ticket-Style Summary

Issue:

A Windows domain environment needed to be created and tested using Windows Server 2019 and Windows 10 virtual machines.

Impact:

Without a domain controller, users, computers, and resources could not be centrally managed.

Steps Taken:

- Verified server and client network connectivity.
- Enabled ICMPv4 traffic for ping testing.
- Renamed the Windows Server 2019 VM to WARA-SERVER.
- Installed Active Directory Domain Services.
- Promoted the server to a domain controller.
- Created a new domain named `wara.edu`.
- Verified AD DS startup in Event Viewer.
- Renamed the Windows 10 VM to WARA-CLIENT.
- Joined the Windows 10 client to the domain.
- Verified the WARA-CLIENT computer object in Active Directory.

Root Cause:

No domain environment had been configured yet.

Resolution:

Installed AD DS, promoted the server to a domain controller, created the domain, and joined the Windows 10 client to the domain.

Status:

Completed successfully.

12. Lessons Learned

In this lab, I learned how Active Directory Domain Services provides centralized management for users, computers, and resources. I also learned why server naming, static IP addressing, DNS, and client connectivity are important before joining a computer to a domain.

This lab helped me understand the foundation of many real IT support tasks, especially domain login troubleshooting, computer account verification, and Windows client support in a business environment.